

From: Sneller, Cindy (DEQ)
Sent: 2/12/2019 10:45:42 AM
To: "Berdinski, Thomas (DEQ)" <BERDINSKIT@michigan.gov>; "Kammer, Stephanie (DEQ)" <KAMMERS@michigan.gov>; "Person, Michael (DEQ)" <PERSONM@michigan.gov>
Cc: "Worm, Michael (DEQ)" <WORMMM@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

That's good to know.
Cindy

From: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Sent: Tuesday, February 12, 2019 10:44 AM
To: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>; Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>; Person, Michael (DEQ) <PERSONM@michigan.gov>
Cc: Worm, Michael (DEQ) <WORMMM@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

Just FYI – Cadillac is comfortable with our thoughts on the results. Jeff Dietlin thinks they will resample the biosolids again on their own, and ensure they get the solids portion analyzed, probably in the spring. Their LF PFOS-source plans to deep well inject their leachate beginning in about 6 months, eliminating the discharge to the WWTP...

I'll be sending out the Bin 2 letter soon, approving their Interim and Summary Reports.

Thanks for the help!

Tom

From: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>
Sent: Tuesday, February 12, 2019 9:47 AM
To: Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>; Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>; Person, Michael (DEQ) <PERSONM@michigan.gov>
Cc: Worm, Michael (DEQ) <WORMMM@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

Here is Bronson's to compare.

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From: Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>
Sent: Tuesday, February 12, 2019 9:39 AM
To: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>; Person, Michael (DEQ) <PERSONM@michigan.gov>; Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

Can you forward the lab report for Cadillac's biosolids and effluent? Thanks.

From: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Sent: Tuesday, February 12, 2019 9:17 AM
To: Person, Michael (DEQ) <PERSONM@michigan.gov>; Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>; Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

I would like to be involved, if possible, because this involves Cadillac and their evaluation of leachate acceptance (their only confirmed source). They believe the results indicate their biosolids are fine. Their effluent results put them in Bin 2 (<WQS). The limited leachate data they have is about 120 ng/l PFOS. I am available at 10 but would need a call in #.

Thanks,
Tom

From: Person, Michael (DEQ) <PERSONM@michigan.gov>
Sent: Tuesday, February 12, 2019 9:10 AM
To: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>; Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>
Cc: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

Great, looks like tom is out but that's Ok, lets try at 10 am?? , Ill just try calling both of you , If I cant do that Ill set up a teleconf. Thanks mike

From: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>
Sent: Tuesday, February 12, 2019 9:06 AM
To: Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>; Person, Michael (DEQ) <PERSONM@michigan.gov>
Cc: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

I'm available all day as well.
Cindy

From: Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>
Sent: Tuesday, February 12, 2019 9:04 AM
To: Person, Michael (DEQ) <PERSONM@michigan.gov>; Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>

Cc: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>

Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

I think a quick call would be helpful. I'm available all day today.

From: Person, Michael (DEQ) <PERSONM@michigan.gov>

Sent: Tuesday, February 12, 2019 9:01 AM

To: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>; Kammer, Stephanie (DEQ) <KAMMERS@michigan.gov>

Cc: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>

Subject: RE: PLEASE look at this - FW: info on % solids & % moisture - Update

Cindy, sorry but I do not feel qualified to try to answer this. My hope was that Test America could advise. I am forwarding to Stephanie, to see perhaps she has a recommendation on a next step here. Should we talk with someone at the deq lab regarding this? Stephanie, I would be happy to host a quick teleconference if you wish to discuss this with us? Thanks mike

Mike Person
Michigan Biosolids Program
personm@michigan.gov
989-297-0779

From: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>

Sent: Tuesday, February 12, 2019 8:04 AM

To: Person, Michael (DEQ) <PERSONM@michigan.gov>

Subject: PLEASE look at this - FW: info on % solids & % moisture - Update

Importance: High

Mike – Please read/look at this. I sent it to you last week. Tom Berdinski is asking for direction on this.

I believe someone in upper management needs to make a decision:

- What kind of results are we going to accept from the facility? What 'units' required? mg/L, ug/L, mg/kg, ug/kg; What form? Water, waste, biosolids;
- If they give us results in the wrong 'units'/form are we going to make them get the results in the correct unit/form?

I don't feel we (as DEQ staff) should have to decipher the results that a facility gives to us. We shouldn't be the ones figuring out the conversion, which is what the laboratory does.

This needs to be decided.

Right now we have the wrong results from Cadillac.

Who is going to tell them they have to get the results in the correct 'unit'/form, when they should've done that from the 'get go'.

Cindy

From: Sneller, Cindy (DEQ)

Sent: Thursday, February 7, 2019 11:54 AM

To: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>; Person, Michael (DEQ) <PERSONM@michigan.gov>

Subject: RE: info on % solids & % moisture - Update

Importance: High

I just got off the phone w/ Test America. Here is what happened:

Test America was overloaded w/ PFAS testing at the end of 2018 and sent some of their tests to a contracted lab - Eurofins. Cadillac's results are from Eurofins.

Test America knows that when biosolids samples come, they test the Total Solids and % moisture, so they can calculate the dry wt after the testing is completed to give the dry wt. (mg/kg) result, in their reports.

Eurofins usually does the same thing.

BUT,

Cadillac *specifically requested* their sample be tested as a wastewater sample, therefore they only provided the result in the liquid form & the % moisture only.

Sue Schafer remembers this one, because Cindy Tomaszewski was adamant that Cadillac get their results this way.

I then asked her about the Total solids for this sampling result and she said it would be *the inverse* of the % moisture.

Then you take the parameter that has a hit and divide it by the Total solids to get the dry wt. calculation.

So, in the case of Cadillac's results: They had a % moisture of 97.3, the Total solids would be 2.7 % or 2.7 ng/l (or 0.0027 ug/l).

Cadillac's numbers:

The PFOS is 190 ng/l, PFOA 130 ng/l and the % moisture is 97.3. (The duplicate is PFOS 220 ng/l, PFOA 140 ng/l).

These need to be converted, which is something the labs do and report as such.

Eurofins would have completed Cadillac's just like they normally would, but Cadillac specifically asked for the results as a water test.

I tried doing the calculation from what Sue told me and something doesn't seem correct, as I end up w/ 70,370 ug/kg on the PFOS.

I say Cadillac needs to get the correct numbers from their lab, which Sue S. (from Test America) said they would do (or their contract lab – Eurofins), but the request has to come from Cadillac.

Thoughts?????

Cindy

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From: Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Sent: Tuesday, January 29, 2019 8:44 AM
To: Person, Michael (DEQ) <PERSONM@michigan.gov>; Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>
Subject: RE: info on % solids & % moisture

Cindy,

Did you ever hear back from Test America?

Thanks,
Tom

From: Person, Michael (DEQ) <PERSONM@michigan.gov>
Sent: Thursday, January 24, 2019 11:55 AM
To: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>; Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Subject: RE: info on % solids & % moisture

Great work Cindy. Thank you. Let us know what you find out. I have a meeting tomorrow with AECO on the biosolids sampling sop and may have a chance to inquire with those guys on this questions.

From: Sneller, Cindy (DEQ) <SNELLERC@michigan.gov>
Sent: Thursday, January 24, 2019 11:17 AM
To: Person, Michael (DEQ) <PERSONM@michigan.gov>; Berdinski, Thomas (DEQ) <BERDINSKIT@michigan.gov>
Subject: info on % solids & % moisture

Ok, so the attachment is a comparison I'm doing on Zeeland's Biosolids results for the sampling issue. Anyway, just want you guys to look at the part of the Total Solids and % Moisture rows. I have a question mark next to the number, sometimes it is the moisture, other times it is the Total Solids. I'm guessing at those numbers

But, in all the results I've viewed for biosolids metals & nutrient results, these seem to turn out that way. Bronson's Test America results prove this.

Attached - Bronson's COC didn't ask for Total Solids and they were provided w/ the Total solids results & % moisture. Why?

Attached - Cadillac's COC didn't ask for Total Solids and they **weren't** provided w/ the Total solids results. But they got one % Moisture result. Why?

I've got a message into the Test America lab & am waiting for them to call me back.

I'll keep you posted.
Cindy

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